



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2024 Sep01: IIT M FN EXAM QDF1

Exam Mode

Learning Mode

View Question Paper Summary

## QUESTION MENU

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## CONTROLS

SUBMIT EXAM

Your Score

**0.00 / 50.00**

(0%)

Question 1 : 640653902368

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL : SEMESTER II: MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET

TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

 Discussions (0)



Question 2 : 640653902369

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Consider the two statements given below:

**Statement-P:** There exists a square matrix of order 2 such that  $A^2 = 0$ ,  $A^T = A$  and  $A \neq 0$ .

**Statement-Q:** There exists a square matrix of order 2 such that  $A^2 = 0$  and  $A \neq 0$ .

OPTIONS :

☐ Statement-P is true, but statement-Q is false.

☐ Statement-P is false, but statement-Q is true.

☐ Both statements are true.

☐ Both statements are false.

Your score : 0

 Discussions (0)



Question 3 : 640653902370

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Consider the following system of equations:

$$2x_1 - x_2 + x_3 = b_1$$

$$x_1 + x_2 - 2x_3 = b_2$$

$$-3x_2 + 5x_3 = b_3$$

Under which of the following conditions is the system consistent?

OPTIONS :

☐  $b_2 = 0$

☐  $b_1 = 2b_2$

☐  $b_3 = b_1 - 2b_2$

☐  $b_1 + b_2 + b_3 = 0$

Your score : 0

Discussions (0)



Question 4 : 640653902371

View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Which of the following is an orthonormal basis for  $\text{span}\{(1, 1, -1), (1, 2, 0)\}$ ?

OPTIONS :

☐  $\left\{ \frac{1}{\sqrt{3}}(1, 1, -1), \frac{1}{\sqrt{2}}(0, 1, 1) \right\}$

☐  $\left\{ \frac{1}{\sqrt{3}}(1, 1, -1), \frac{1}{\sqrt{5}}(1, 2, 0) \right\}$

☐  $\left\{ \frac{1}{\sqrt{2}}(1, 0, 1), \frac{1}{\sqrt{2}}(-1, 0, 1) \right\}$

☐  $\left\{ \frac{1}{\sqrt{2}}(0, 1, 1), \frac{1}{\sqrt{2}}(0, -1, 1) \right\}$

Your score : 0

Discussions (0)



## Question 5 : 640653902372

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

$$\dim(\text{Span}(S)) = 5 \leq V$$

Let  $S = \{u_1, u_2, u_3, u_4, u_5\}$  be a linearly independent subset of a vector space  $V$ . Select all true statements.

OPTIONS :

☐ There exists a basis  $B$  for  $V$  such that  $S \subseteq B$ 
☐  $\dim(V) \geq 5$ 
☐  $\text{span}(S) = V$ 
☐  $S$  is a subset of every basis  $B$ 

Your score : 0

Discussions (0)



## Question 6 : 640653902373

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the function  $f$ :

$$f(x, y) = \begin{cases} \frac{y^4}{x^2 + y^2} & , (x, y) \neq (0, 0) \\ 0 & , (x, y) = (0, 0) \end{cases}$$

If  $f_x, f_y$  denote the partial derivatives, select all true statements from the options given below.

OPTIONS :

☐  $f_x(0, 0)$  does not exist.

☐  $f_y(0, 0)$  does not exist.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{h^4}{h^2+0} - 0}{h} = \lim_{h \rightarrow 0} \frac{h^4}{h^3} = \lim_{h \rightarrow 0} h = 0 \checkmark$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, 0+h) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{h^4}{0^2+h^2} + 0}{h} = \lim_{h \rightarrow 0} \frac{h^4}{h^3} = 0$$

☐  $f_x(0,0) = 0$

☐  $f_y(0,0) = 0$

Your score : 0

Discussions (0)



### Question 7 : 640653902374

View Solutions (0)

Total Mark : 3.00 | Type : SA

Find the dimension of the affine subspace corresponding to the set of all solutions to the following system: 

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



### Question 8 : 640653902375

View Solutions (0)

Total Mark : 3.00 | Type : SA

Let  $v = (\sqrt{2}, -\sqrt{2}, 6) \in \mathbb{R}^3$ .  $(a, b, c)$  is the vector obtained from  $v$  after rotating the XY-plane anti-clockwise by  $45^\circ$  about the Z-axis. Find the value of  $a+b+c$ . Enter the nearest integer as your answer. 

Answer (Numeric):

Answer

Accepted Answer : 8

Your score : 0

Discussions (0)



## Question 9 : 640653902376

View Solutions (0)

Total Mark : 3.00 | Type : SA

Evaluate  $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(4x^2 + 4y^2)}{x^2 + y^2}$ .



$$\lim_{r^2 \rightarrow 0} \frac{\sin(4r^2)}{(4r^2)} = 4 \lim_{r^2 \rightarrow 0} \frac{\sin(4r^2)}{(4r^2)} = 4$$

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)



## Question 10 : 640653902377

View Solutions (0)

Total Mark : 3.00 | Type : SA

Consider the following function:



$$f(x, y) = \begin{cases} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}, & x \neq y \text{ and } x \geq 0 \text{ and } y \geq 0 \\ c, & \text{otherwise} \end{cases}$$

For what value of  $c$  is the function  $f$  continuous at  $(0,0)$ ?

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)



## Question 11 : 640653902378

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

If the directional derivative of  $f(x, y) = x^3y + xy^2 + 8$  at the point  $(1, 2)$  in the direction  $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$  is  $\frac{p}{\sqrt{2}}$ , find the value of  $p$ . 

Answer (Numeric):

Accepted Answer : 5

Your score : 0

[Discussions \(0\)](#)

## Question 12 : 640653902379

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

The equation of the tangent plane to the function  $f(x, y) = e^x(x + y)$  at the point  $(1, 0)$  is given by  $Ax + By + C = z$ . Find the value of  $\frac{A + B + C}{e}$ . 

Answer (Numeric):

Accepted Answer : 2

Your score : 0

[Discussions \(0\)](#)

## Question 13 : 640653902380

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

If  $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 5 \\ -4 & 9 & 5 \end{bmatrix}$ , find the determinant of  $A^3 - 3A^2 + 2A$ . 

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)



## Question 14 : 640653902392

Total Mark : 0.00 | Type : COMPREHENSION

Consider a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ :

$$T(x, y, z) = (x - y + z, x + z, y - 2z)$$

Use the given standard basis for both domain and co-domain for matrix representation ,  
Answer the sub-questions

Your score : 0

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

matrix mult

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y + z \\ x + z \\ y - 2z \end{pmatrix}$$

Question 15 :  
640653902393

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the matrix representation of  $T$ ? 

OPTIONS :

☐  $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

if  $\Delta \neq 0$  full rank 0 nullity $\Delta \rightarrow$  rank nullity theoremwhat  $\Delta = 2 \neq 0$  full rank, nullity is 0  
nullspace is singleton



☐  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Your score : 0

Discussions (0)



**Question 16 :**  
**640653902394**

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MSQ

Select all true statements.

OPTIONS :

☐ The kernel of  $T$  is  $\{(0, 0, 0)\}$

☐ The kernel of  $T$  is  $\mathbb{R}^3$

☐ The image of  $T$  is  $\{(0, 0, 0)\}$

☐ The image of  $T$  is  $\mathbb{R}^3$

Your score : 0

Discussions (0)



**Question 17 :**  
**640653902395**

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

$T^{-1}$  is a linear transformation that is the inverse of  $T$ . That is, if

$$T(x, y, z) = (a, b, c), \text{ then}$$

$$T^{-1}(a, b, c) = (x, y, z).$$

Find the matrix representation of  $T^{-1}$ ,

$\perp \text{adj}(A)^T$   
 $|\Delta|$

OPTIONS :

☐  $\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 \\ -2 & 2 & 0 \\ -1 & 1 & -1 \end{bmatrix}$

☐  $\frac{1}{2} \begin{bmatrix} 1 & -2 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} -1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

Your score : 0

Discussions (0)



### Question 18 : 640653902381

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let  $f : D \rightarrow \mathbb{R}$  be defined as follows:

$f(x, y) = \sqrt{1 - x^2 - y^2}$

where  $D \subset \mathbb{R}^2$  is the domain of  $f$ .

Your score : 0

$\sqrt{x-1} \rightarrow x \geq 1$   
 complex no.

$1 - x^2 - y^2 \geq 0$   
 $1 \geq x^2 + y^2$

$x^2 + y^2 \leq 1$



## Question 19 :

640653902382

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

Choose the largest possible D that would be a valid domain from the options given below.

OPTIONS :

- ☐  $\mathbb{R}^2$ 
☐  $\{(x, y) : |x| \leq 1 \text{ and } |y| \leq 1; x, y \in \mathbb{R}\}$ 
☐  $\mathbb{R}^2 \setminus \{(x, y) : x^2 + y^2 < 1; x, y \in \mathbb{R}\}$ 
☐  $\mathbb{R}^2 \setminus \{(x, y) : x^2 + y^2 \leq 1; x, y \in \mathbb{R}\}$ 
☒  $\{(x, y) : x^2 + y^2 \leq 1; x, y \in \mathbb{R}\}$

Your score : 0

[Discussions \(0\)](#)


## Question 20 :

640653902383

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Which of the following is the range of  $f$  using the domain chosen from the previous question? Recall that  $[a, b] = \{x : a \leq x \leq b; x \in \mathbb{R}\}$

OPTIONS :

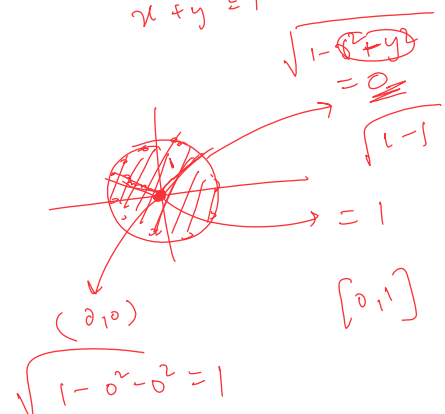
- ☒  $[0, 1]$ 
☐  $[-1, 1]$

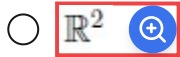


Domain

$$x^2 + y^2 = 1$$

$$x^2 + y^2 \leq 1$$





Your score : 0

Discussions (0)

**Question 21 : 640653902384**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function  $f(x, y) = x^2 + xy + y^2 - 7x - 8y$ . 

Your score : 0

**Question 22 :  
640653902385** View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

If  $f$  has a critical point at  $(a, b)$ ,   
find  $a + b$ .

Answer (Numeric):

Answer

Accepted Answer : 5

Your score : 0

Discussions (0)

**Question 23 :  
640653902386** View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the determinant of the   
Hessian at  $(a, b)$ .

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

 Discussions (0)



Question 24 :  
640653902387

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the nature of the critical point?

OPTIONS :

☐ Local maximum

☐ Local minimum

☐ Saddle point

Your score : 0

 Discussions (0)



Question 25 : 640653902388

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Find three positive numbers  $x, y, z$  whose sum is 10 such that  $x^2 y z^2$  is maximum. 

Your score : 0

$x + y + z = 10$   
 $y = 10 - x - z$

$\uparrow f(x, z) = 10x^2 z^2 - x^3 z^2 - x^2 z^3$

$x^2 (10 - x - z) z^2$   
 $x^2 z^2 (10 - x - z)$



**Question 26 :**  
**640653902389**

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Enter the value of  $x$ . 

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

$$V = \left( \frac{20xz^2 - 3x^2z^2 - 2xz^3}{x \cdot y}, \frac{20x^2z - 2x^3z - 3x^2z^2}{x^2z} \right)$$

$$xz(20z - 3xz - 2z^2) \quad x^2z(20 - 2x - 3z)$$

$$xz^2(20 - 3x - 2z) \quad x=0 \rightarrow z=0$$

$$xz^2(20 - 3x - 2z) \quad 3 \times 2x + 3z = 20$$

$$x=0 \text{ or } z=0 \quad 2 \times 3x + 2z = 20$$

[Discussions \(0\)](#)


**Question 27 :**  
**640653902390**

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Enter the value of  $y$ . 

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

$$3x + 8 = 20$$

$$3x = 12$$

$$x = 4$$

[Discussions \(0\)](#)


**Question 28 :**  
**640653902391**

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Enter the value of  $z$ . 

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

 Discussions (0)



 **SUBMIT EXAM**